

Package ‘BayGMST’

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Type: Package
Title: Bayesian Reconstruction of Global Mean Surface Temperature
Version: 0.1.0
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License: GPL (≥ 3)

READ THIS FIRST

This is a **hand-typeset preview**, not the real, authoritative CRAN reference manual. It was produced in a sandbox with **no R installation available**, so it could not be generated the normal way (`devtools::document()` to build `man/*.Rd`, then `devtools::build_manual()` / R CMD Rd2pdf to render this PDF from those `.Rd` files). Content below was written by hand to mirror that output as closely as possible, from the same source descriptions as the `roxygen2` comments in R/.

Before treating this as final: run the two commands above (see `CRAN-READINESS.md` at the package root for the full checklist) and let the real output supersede this file.

Text marked **[GAP]** like [this](#) is a genuine information gap I could not resolve myself. Text marked **[FLAG]** like [this](#) is a decision, judgment call, or unresolved issue from the original code that needs Tyler’s or Julien’s sign-off, not something I decided unilaterally.

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1 R topics documented:

BayGMST-package	BayGMST: Bayesian Reconstruction of Global Mean Surface Temperature
cv_baygmst	K-fold cross-validate the BayGMST model over the instrumental period
fit_baygmst	Fit the BayGMST Bayesian hierarchical AR(1) reconstruction model
plot_cv	Plot k-fold cross-validation reconstructions
plot_posterior_densities	Posterior density plots for a fitted BayGMST model
plot_reconstruction	Plot a BayGMST reconstruction
plot_trace	Trace plots for a fitted BayGMST model
print.baygmst_cv	Print method for "baygmst_cv" objects
print.baygmst_fit	Print a fitted BayGMST model
print.summary.baygmst_fit	Print method for "summary.baygmst_fit" objects
reconstruct	Extract a tidy reconstruction from a fitted BayGMST model
reduce_proxies	Reduce a multi-proxy network to a single representative proxy series
summary.baygmst_fit	Summarize a fitted BayGMST model
transform_forcings	Transform raw radiative forcings for use with <code>fit_baygmst()</code>

2 BayGMST-package

BayGMST: Bayesian Reconstruction of Global Mean Surface Temperature

Description

Fits a Bayesian hierarchical AR(1) state-space model relating global mean surface temperature (GMST) to a reduced paleoclimate proxy network, instrumental temperature observations, and radiative forcings (greenhouse gases, volcanic aerosols, and solar irradiance). Missing pre-instrumental temperatures are estimated jointly with the model parameters using Stan via the `instantiate` and `cmdstanr` packages.

Model fitting requires CmdStan

`fit_baygmst()` and `cv_baygmst()` compile and sample a Stan model, which requires a working CmdStan installation. Check with `instantiate::stan_cmdstan_exists()` first. `reduce_proxies()` and `transform_forcings()` do not require CmdStan.

Provenance

Restructured from a collection of top-level analysis scripts (see `NEWS.md` for exactly what changed, and `inst/legacy-scripts/` for the untouched originals). **[GAP]** The **Description:** field's exact public-facing wording in `DESCRIPTION` was drafted by me from the `README` and the thesis proposal's Project II motivation section — it has not been reviewed or approved by Tyler or Julien. Please read and edit it before any submission.

Author(s)

Tyler Bagwell <teb6@rice.edu> (maintainer), Julien Emile-Geay <julieneg@usc.edu>. **[FLAG]** These are the only two contributors visible in `git log`; add others if appropriate.

3 transform_forcings

Transform raw radiative forcings for use with `fit_baygmst()`

Description

Applies the forcing transformations used throughout the BayGMST pipeline: volcanic forcing is mapped to a negative saturating form, greenhouse-gas (CO₂) forcing is log-transformed relative to a reference concentration, and solar forcing is centered on its own mean. Ported from the normalization block near the top of the original `BayGMST_v1.0.R` script.

Usage

```
transform_forcings(G, V, S, co2_c0 = 280, co2_coef = 5.35, vol_coef = 25)
```

Arguments

G	Numeric vector of raw greenhouse-gas forcing (e.g. atmospheric CO ₂ concentration in ppm).
V	Numeric vector of raw volcanic forcing (e.g. aerosol optical depth). Same length as G .
S	Numeric vector of raw solar forcing (e.g. total solar irradiance). Same length as G .
co2_c0	Reference/baseline CO ₂ concentration used in the log transform. Default 280 (ppm).
co2_coef	Multiplier applied to the log CO ₂ ratio. Default 5.35.
vol_coef	Multiplier applied to the volcanic saturating transform. Default 25.

Details

$$V' = -|\text{vol_coef}| \times (1 - e^{-V}), \quad G' = \text{co2_coef} \times \log(G/\text{co2_c0}), \quad S' = S - \bar{S}$$

Value

A list with three numeric vectors, **G**, **V**, and **S** – the transformed forcings.

Examples

```
fc <- transform_forcings(
  G = c(280, 300, 340, 400),
  V = c(0, 0.05, 0.02, 0.10),
  S = c(1360, 1361, 1360.5, 1361.2)
)
fc$G
```

4 reduce_proxies

Reduce a multi-proxy network to a single representative proxy series

Description

Collapses a matrix of paleoclimate proxy records into a "representative proxy" series suitable for `fit_baygmst()`, following the segmented-composite approach of Barboza et al. (2019), ported from `utils/PAGES2k_reducedProxy_UNSC.R`. The proxy network is broken into overlapping time segments of `chunk` years; within each segment, available proxies are regressed against instrumental temperature during `calib_years` using the chosen `method`. The final composite series averages across all segments covering a given year.

Usage

```
reduce_proxies(proxy_matrix, years, temp_calib, calib_years,
method = c("PCR", "LASSO", "sPCR", "SPLS", "SIR"),
chunk = 250, target_adj_r2 = 0.7, max_na_frac = 0.05,
n_components = 3, spls_eta = seq(0.1, 0.9, 0.1))
```

Arguments

<code>proxy_matrix</code>	Numeric matrix/data.frame, one row per year (matching <code>years</code>), one column per proxy. No year column.
<code>years</code>	Integer vector of calendar years, one per row of <code>proxy_matrix</code> .
<code>temp_calib</code>	Numeric vector, instrumental temperature during <code>calib_years</code> .
<code>calib_years</code>	Integer vector, calibration years (subset of <code>years</code> ; <code>length(calib_years) == length(temp_calib)</code>).
<code>method</code>	One of "PCR" (default), "LASSO", "sPCR", "SPLS", or "SIR". "SIR" is experimental — see Details.
<code>chunk</code>	Segment length in years. Default 250. Every segment must fully contain <code>calib_years</code> , or <code>reduce_proxies()</code> errors (see Details).
<code>target_adj_r2</code>	For "PCR": adjusted- R^2 stopping threshold for adding principal components. Default 0.70.
<code>max_na_frac</code>	Proxies missing more than this fraction of a segment are dropped from it. Default 0.05.
<code>n_components</code>	For "sPCR" only: number of supervised components (scalar or per-segment vector). Default 3.
<code>spls_eta</code>	For "SPLS" only: sparsity grid passed to <code>spls::cv.spls()</code> .

Details

Method support. "PCR", "LASSO", and "SPLS" are fully adaptive and supported. "sPCR" is supported but requires `n_components` to be chosen by the caller (not cross-validated, matching the original script).

[FLAG] "SIR" is a simplified, *unvalidated* reimplementation using a single sliced-inverse-regression direction from `dr`, with no per-segment variable pre-selection. It intentionally does *not* reproduce the original script's SIR branch, which called an undocumented `edrSelec()` step from the non-CRAN package 'edr-GraphicalTools' that the original author flagged as never fully working ("NEED TO FIGURE OUT HOW TO INSTALL `edrGraphicalTools`!!"). `config.yml` in the original repository likewise documented SIR as "still under construction." Treat `method = "SIR"` output with corresponding caution.

[FLAG] The segmentation/calibration-window constraint (`chunk` must be small enough that even the shortest segment still spans all of `calib_years`) mirrors an explicit `stop()` check in the original script. I ported

that check explicitly rather than letting a mismatched-length input fail with a cryptic error deep inside `lm()` — but the underlying design constraint itself is inherited from the original code, not something I introduced.

Required packages: "LASSO" needs `glmnet`; "sPCR" needs `superpc`; "SPLS" needs `spls`; "SIR" needs `dr`. All are **Suggests**, only required for the method actually called.

Value

A list with **segments** (data.frame, one column per segment), **composite** (data.frame with columns **year** and **RP1**, the series to pass to `fit_baygmst()`), and **method**.

5 fit_baygmst

Fit the BayGMST Bayesian hierarchical AR(1) reconstruction model

Description

Prepares the data list expected by the `baygmst` Stan model and samples from it with `cmdstanr`, via `instantiate`. Ported from the data-preparation and `mod$sample()` block of the original `BayGMST_v1.0.R` script, restructured to take explicit arguments instead of reading `config.yml` and hardcoded relative paths, and to return an object rather than leaving variables in the global environment.

Usage

```
fit_baygmst(proxy, instrumental_T, forcing_G, forcing_V, forcing_S, years,
  chains = 4, parallel_chains = 2, iter_warmup = 500,
  iter_sampling = 1500, seed = NULL, ...)
```

Arguments

<code>proxy</code>	Representative proxy series – see <code>reduce_proxies()</code> . No missing values.
<code>instrumental_T</code>	Instrumental temperature, NA for pre-instrumental years to reconstruct.
<code>forcing_G</code> , <code>forcing_V</code> , <code>forcing_S</code>	Already-transformed forcings – see <code>transform_forcings()</code> . No missing values.
<code>years</code>	Calendar years, defining row order for all arguments above.
<code>chains</code> , <code>parallel_chains</code> , <code>iter_warmup</code> , <code>iter_sampling</code> , <code>seed</code>	Passed to <code>cmdstanr</code> 's <code>\$sample()</code> . <code>iter_sampling</code> $\geq$ 1000 required.
<code>...</code>	Additional arguments passed to <code>\$sample()</code> .

Value

An object of class "baygmst_fit": a list with `fit` (the `CmdStanMCMC` object), `years`, `idx_obs`, `idx_mis`, `data`, and `call`.

Note

[GAP] This function does not write any files by default (no `auto-ggsave()`, no auto-CSV export) – a deliberate deviation from the original script, which wrote directly to `cfg$folder_paths$...`. This matches CRAN policy on functions not writing to the filesystem by default, but it does mean any code that relied on those side-effect files (e.g. `outputs/reconstructions/fit_post_summaries.csv`) needs to be updated to call `write.csv()` itself on the returned object.

6 `cv_baygmst`

K-fold cross-validate the BayGMST model over the instrumental period

Description

Partitions the instrumental years of `instrumental_T` into `nfold` contiguous blocks; for each, refits the model with that block hidden, and reports recovery accuracy. Ported from the fold loop in `cv_v0.1.R`.

Usage

```
cv_baygmst(proxy, instrumental_T, forcing_G, forcing_V, forcing_S, years,
  nfold = 3, chains = 4, parallel_chains = 2, iter_warmup = 500,
  iter_sampling = 1500, seed = NULL, ...)
```

Arguments

Same as `fit_baygmst()`, plus `nfold` (number of folds, default 3).

Value

An object of class `"baygmst_cv"`: `folds` (combined data.frame of held-out predictions), `r2` and `mse` (per-fold, posterior mean of the Stan-computed `r2_cv/mse_cv` generated quantities), `years`, `call`.

Note

[FLAG] Two blocks in the original `cv_v0.1.R` were dropped, not ported: (1) a `Forcings.projections/RCP-scenario` block that was built but never referenced again (dead code), and which additionally hardcoded a path to a specific contributor's machine (`/Users/tylerbagwell/...`); and (2) a large commented "OLD" duplicate of the fold loop at the end of the file. Confirm nothing downstream still depended on either.

[FLAG] This implementation reports the fully Bayesian posterior mean of the Stan model's own `r2_cv/mse_cv` generated quantities. The original (non-dead) R loop instead computed a plug-in point-estimate R^2 in R from the posterior mean reconstruction. The two will be close but not identical – flagging the substitution explicitly rather than silently changing what "R2" means.

7 reconstruct

Extract a tidy reconstruction from a fitted BayGMST model

Description

Builds a tidy data.frame combining observed instrumental temperatures with the posterior reconstruction of pre-instrumental years – equivalent to the `gmst_reconstruction_data.csv` the original script wrote to disk as a side effect, but returned as an object instead.

Usage

```
reconstruct(fit, probs_inner = c(0.16, 0.84), probs_outer = c(0.025, 0.975))
```

Value

A data.frame with columns `year`, `type` ("instrumental"/"reconstruction"), `T_obs`, `T_mean`, `T_lo_inner`, `T_hi_inner`, `T_lo_outer`, `T_hi_outer`.

8 `plot_reconstruction`, `plot_trace`, `plot_posterior_densities`, `plot_cv`

Plotting functions

Description

- `plot_reconstruction(fit, title = "GMST Reconstruction")` – ported from the `p_ts` block of BayGMST_v1.0.R.
- `plot_trace(fit, parameters = ...)` – ported from the trace-plot block.
- `plot_posterior_densities(fit, parameters = ...)` – ported from the posterior-histogram block.
- `plot_cv(cv)` – ported from the `p_ts_cv` block of `cv_v0.1.R`.

All four return a `ggplot` object instead of calling `ggsave()`/auto-printing as a side effect; save with `ggplot2::ggsave()` if a file is wanted. **[FLAG]** The original script combined the reconstruction and posterior-density panels into one figure via `patchwork (p_ts + p_hist)`. Splitting these into separate functions means that combination is no longer automatic – users who want the original combined layout can do `plot_reconstruction(fit) | plot_posterior_densities(fit)` themselves via `patchwork` (moved to `Suggests`, since the package itself no longer calls it directly).

9 print/summary methods

S3 methods

Description

- `print.baygmst_fit(x, ...)` – one-line-per-field overview (year range, chains/draws, call).
- `summary.baygmst_fit(object, ...)` – posterior parameter summary (`alpha0`, `alpha1`, `phi_R`, `phi_T`, `beta0`, `betaG`, `betaS`, `betaV`, `sigma_y`, `sigma_z`) plus instrumental-period fit performance (RMSE, MAE, Bias, Correlation, R^2 , detrended R^2), replacing the `perf_stats/R2_dt` block of `BayGMST_v1.0.R`.
- `print.summary.baygmst_fit(x, ...)` – pretty-prints the above.
- `print.baygmst_cv(x, ...)` – per-fold and median R^2 .

Known limitations index

[GAP] The unresolved possible bug in `src/stan/baygmst.stan` (`y_ins_fitted` block, `t == 1` branch: `z[NT_mis]` used as an index where `z[idx_mis[NT_mis]]` looks more likely correct) is carried forward *unchanged* from the original code. This restructuring does not alter model math without Tyler/Julien's statistical sign-off. See `NEWS.md` for the exact location and reasoning.

[FLAG] Several `library()` calls in the original scripts (`car`, `fda`, `ggmap`, `maps`) had no evident corresponding usage in the code as written and were not carried into `DESCRIPTION`. If a hidden use is found, re-add them.

[GAP] `DESCRIPTION`'s `RoxygenNote` field is a placeholder (7.3.2) – `devtools::document()` will set the real value.

[GAP] `NAMESPACE` and every file in `man/` are hand-authored placeholders (see the header comment in each), not actual `roxygen2` output, because this restructuring was done without an R installation available to run it. `CRAN-READINESS.md` has the exact commands to regenerate and verify them for real.